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Safe motion planning for machinery operation

Abstract

A method of safely operating machinery in a workspace includes recording images of a portion of a workspace. The method also includes generating a three-dimensional (3D) representation of the portion of the workspace based on the recorded images, where the 3D representation includes one or more volumes that correspond to the portion of the workspace. Additionally, the method includes identifying one or more of the volumes as being either occupied or unoccupied. Further, the method includes mapping one or more safe zones based on the one or more identified volumes, where the safe zones correspond to one or more regions within the portion of the workspace for safe operation of machinery.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 18/479,933 (filed Oct. 3, 2023), which is a continuation-in-part of U.S. patent application Ser. No. 17/101,622 (filed Nov. 23, 2020), which is a continuation of U.S. patent application Ser. No. 16/129,999 (filed Sep. 13, 2018), which is a continuation of U.S. patent application Ser. No. 15/889,523 (filed Feb. 6, 2018). The present application also claims priority to U.S. Provisional App. No. 62/455,828 (filed Feb. 7, 2017) and U.S. Provisional App. No. 62/455,834 (filed Feb. 7, 2017). Each of these applications is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) The field of the invention relates, generally, to ensuring safe operation of industrial machinery, and in particular to systems and methods for detecting safe and/or unsafe conditions in a monitored workspace and, based thereon, planning safety-constrained motion of the machinery.

BACKGROUND

(2) Industrial machinery is often dangerous to humans. Some machinery is dangerous unless it is completely shut down, while other machinery may have a variety of operating states, some of which are hazardous and some of which are not. In some cases, the degree of hazard may depend on the location or distance of the human with respect to the machinery. As a result, many “guarding” approaches have been developed to separate humans and machines and to prevent machinery from causing harm to humans. One very simple and common type of guarding is simply a cage that surrounds the machinery, configured such that opening the door of the cage causes an electrical circuit to place the machinery in a safe state. If the door is placed sufficiently far from the machinery to ensure that the human can't reach it before it shuts down, this ensures that humans can never approach the machinery while it is operating. Of course, this prevents all interaction between human and machine, and severely constrains use of the workspace.

(3) More sophisticated types of guarding may involve, for example, optical sensors. Examples include light curtains that determine if any object has intruded into a region monitored by one or more light emitters and detectors, and 2D LIDAR sensors that use active optical sensing to detect the minimum distance to an obstacle along a series of rays emanating from the sensor, and thus can be configured to detect either proximity or intrusion into pre-configured two-dimensional (2D) zones. More recently, systems have begun to employ three-dimensional (3D) depth information using, for example, 3D time-of-flight cameras, 3D LIDAR, and stereo vision cameras. These sensors offer the ability to detect and locate intrusions into the area surrounding industrial machinery in 3D, which has several advantages. For example, a 2D LIDAR system guarding an industrial robot will have to stop the robot when an intrusion is detected well beyond an arm's-length distance away from the robot, because if the intrusion represents a person's legs, that person's arms could be much closer and would be undetectable by the 2D LIDAR. However, a 3D system can allow the robot to continue to operate until the person actually stretches his or her arm towards the robot. This allows a much tighter interlock between the actions of the machine and the actions of the human, which facilitates many applications and saves space on the factory floor, which is always at a premium. Additionally, in complex workcells it can be very difficult to determine a combination of 2D planes that effectively monitors the entire space; 3D sensors properly configured, can alleviate this issue.

(4) Because human safety is at stake, guarding equipment typically employed in stringent industry standards may specify failure rates for hardware components and rigorous development practices for both hardware and software components. Standards-compliant systems must ensure that dangerous conditions can be detected with very high probability, that failures of the system itself are detected, and that the system responds to detected failures by transitioning the equipment being controlled to a safe state. Simply keeping humans and machines apart represents a far simpler guarding task than detecting unsafe conditions when humans actively work with machines that can injure them.

(5) Task performance by a robot generally involves some degree of motion planning, i.e., finding a collision-free path from a robot's starting position and/or state to a goal position and/or state. Most simply, motion planning provides a trajectory along which the robot can move its linked arm to reach a desired object without colliding with itself or other objects. Introducing the possibility of humans moving into and out of proximity to the robot substantially complicates motion planning, since, as noted, safety standards are stringent and human movements are not always predictable. An optimized sequence of trajectories for performing a task may become suboptimal if interrupted by movement of humans and other "dynamic obstacles." It may be necessary to stop or slow the robot temporarily or recompute one or more trajectories altogether. Accordingly, a need exists for motion-planning methodologies that incorporate safety constraints and knowledge of moving humans and objects in a workspace. Ideally, such methodologies would maximize efficient use of the controlled machinery while adhering to rigorous safety standards.

SUMMARY

(6) Embodiments of the present invention provide systems and methods for monitoring a workspace for safety purposes using sensors distributed about the workspace. The workspace may contain one or more pieces of equipment that can be dangerous to humans, for example, an industrial robot and auxiliary equipment such as parts feeders, rails, clamps, or other machines. The sensors are registered with respect to each other, and this registration is monitored over time. Occluded space as well as occupied space is identified, and this mapping is frequently updated.

(7) Regions within the monitored space may be marked as occupied, unoccupied or unknown; only empty space can ultimately be considered safe, and only when any additional safety criteria—e.g., minimum distance from a piece of controlled machinery—is satisfied. In general, raw data from each sensor is analyzed to determine whether, throughout the zone of coverage corresponding to the sensor, an object or boundary of the 3D mapped space has been definitively detected.

(8) In various embodiments, once the safe zone(s) is identified using the sensor data, a constrained motion plan for the machinery specifying a trajectory and/or a state (e.g., an orientation, a pose, a velocity, etc.) of the machinery from the commencement to the end of a task can be computationally created. Based on the constrained motion plan, the machinery may perform the task without entering the identified unsafe zone(s)—i.e., the machinery and its appendages are constrained to operate within the unoccupied space or space explicitly defined as safe. In one implementation, the motion plan includes multiple trajectories; each trajectory may be assigned with a cost value based on, for example, the length of the trajectory and/or an operation time of the machinery on the trajectory. The trajectory corresponding to the minimal cost value (e.g., the shortest length and/or shortest operation time) and satisfying the safety constraint may be elected, and the machinery can then perform the task in accordance with the selected trajectory. This approach maximizes efficient operation of the machinery within the constraints of safety.

(9) As a person moves within a 3D space, he or she will typically occlude some areas from some sensors, resulting in areas of space that are temporarily unknown. Additionally, moving machinery such as an industrial robot arm can also temporarily occlude some areas. When the person or machinery moves to a different location, one or more sensors will once again be able to observe the unknown space and return it to the confirmed-empty state and therefore safe for the robot or machine to operate in this space. Accordingly, in some embodiments, space may also be classified as "potentially occupied." Unknown space is considered potentially occupied when a condition arises where unknown space could be occupied. This could occur when unknown space is adjacent to entry points to the workspace or if unknown space is adjacent to occupied or potentially occupied space. The potentially occupied space "infects" unknown space at a rate that is representative of a human moving through the workspace. Potentially occupied space stays potentially occupied until it is observed to be empty. For safety purposes, potentially occupied space is treated the same as occupied space.

(10) For some sensor modalities such as those relying on an active optical signal, the ability of a sensor to definitively detect an object or boundary falls off rapidly with distance; that is, beyond a certain distance, a sensor may not be capable of distinguishing between an object and empty space, since the associated illumination levels are too similar. Points or regions at such locations are marked as "unknown" with respect to the relevant sensor, and regions so marked cannot be confirmed as empty by that sensor.

(11) In some embodiments, the motion plan is dynamically adjusted based on the real-time monitoring of the workspace by the sensors. For example, as the person and/or machinery move within the 3D workspace, the safe zone(s) corresponding to the unoccupied space may be updated to avoid colliding with the person and/or machinery. As a result, the trajectory (or trajectories) of the machinery in the motion plan may be adjusted based on the updated safe zone(s) so as to ensure safe and efficient operation of the machinery.

(12) Accordingly, in one aspect, the invention pertains to a system for safely operating machinery in a three-dimensional workspace. In various embodiments, the system includes multiple sensors (e.g., time-of-flight sensors, 3D LIDAR sensors and/or stereo vision cameras) distributed about the workspace, each of the sensors being associated with a grid of pixels for recording images of a portion of the workspace within a sensor field of view, the workspace portions partially overlapping with each other; and a controller configured to computationally generate a three-dimensional representation of the workspace as multiple 3D volumes based on the recorded images; for each of the sensors, identify volumes within a line of sight of the sensor as occupied or unoccupied; map one or more safe volumetric zones within the workspace based on the identified volumes; receive a task specification; and computationally generate a constrained motion plan including one or more trajectories of the machinery to perform the task within the safe volumetric zone(s). In one implementation, the controller is further configured to operate the machinery in accordance with the motion plan.

(13) In various embodiments, the motion plan includes multiple trajectories of the machinery; the controller is further configured to assign a cost value to each of the trajectories; select one of the trajectories based at least in part on the assigned cost values to the trajectories; and cause the machinery to execute the selected trajectory. In one embodiment, the controller is further configured to assign the cost value to each of the trajectories based at least in part on a length of the trajectory, an operation time of the machinery on the trajectory, and/or allowed kinematics of the trajectory. In addition, the controller may be further configured to associate the safe volumetric zone(s) with a safety criterion. In one embodiment, the safety criterion corresponds to a minimum distance from a piece of the controlled machinery.

(14) The safe volumetric zone(s) may include only the unoccupied volumes that have been classified as unoccupied based on two or more of the sensors. In some embodiments, the constrained motion plan precludes entry into a volumetric zone outside the safe volumetric zone(s) by the machinery and any workpiece associated with the machinery or the task. In addition, the controller is responsive to real-time monitoring of the workspace by the sensors and is further configured to alter the motion plan in response to a change in the safe volumetric zone(s). In various embodiments, the controller is further configured to, upon detecting an occlusion in a line-of-sight ray path, (i) classify the volumes associated with the occlusion as occupied volumes, and (ii) classify the volumes between said each of the sensors and the occupied volumes in the line-of-sight ray path as the unoccupied volumes.

(15) In another aspect, the invention relates to a method of safely operating machinery in a three-dimensional workspace. In various embodiments, the method includes the steps of monitoring the workspace with multiple sensors (e.g., time-of-flight sensors, 3D LIDAR sensors and/or stereo vision cameras) distributed thereabout, each of the sensors being associated with a grid of pixels for recording images of a portion of the workspace within a sensor field of view, the workspace portions partially overlapping with each other; based on the recorded images, computationally generating a three-dimensional representation of the workspace as multiple 3D volumes stored in a computer memory; for each of the sensors, identifying volumes within a line of sight of the sensor as occupied or unoccupied; mapping one or more safe volumetric zones within the workspace based on the

identified volumes; receiving a task specification and; computationally generating a constrained motion plan including one or more trajectories of the machinery to perform a specified task within the safe volumetric zone(s). In one implementation, the method further includes operating the machinery in accordance with the motion plan.

(16) In various embodiments, the motion plan includes multiple trajectories of the machinery; the method further includes assigning a cost value to each of the trajectories; selecting one of the trajectories based at least in part on the assigned cost values to the trajectories; and causing the machinery to execute the selected trajectory. In one embodiment, the cost value associated with each of the trajectories is assigned based at least in part on a length of the trajectory, an operation time of the machinery on the trajectory and/or allowed kinematics of the trajectory. In addition, the method may further include associating the safe volumetric zone(s) with a safety criterion. In one embodiment, the safety criterion corresponds to a minimum distance from a piece of the controlled machinery.

(17) The safe volumetric zone(s) may include only the unoccupied volumes that have been classified as unoccupied based on two or more of the sensors. In some embodiments, the constrained motion plan precludes entry into a volumetric zone outside the safe volumetric zone(s) by the machinery and any workpiece associated with the machinery or the task. In addition, the method further includes real-time monitoring the workspace and altering the motion plan in response to a change in the safe volumetric zone(s). The identification of the machinery element and/or workpiece may be performed using a neural network. In various embodiments, the method further includes, upon detecting an occlusion in a line-of-sight ray path, (i) classifying the volumes associated with the occlusion as occupied volumes, and (ii) classifying the volumes between said each of the sensors and the occupied volumes in the line-of-sight ray path as the unoccupied volumes.

(18) In general, as used herein, the term “substantially” means $\pm 10\%$, and in some embodiments, $\pm 5\%$. In addition, reference throughout this specification to “one example,” “an example,” “one embodiment,” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with the example is included in at least one example of the present technology. Thus, the occurrences of the phrases “in one example,” “in an example,” “one embodiment,” or “an embodiment” in various places throughout this specification are not necessarily all referring to the same example. Furthermore, the particular features, structures, routines, steps, or characteristics may be combined in any suitable manner in one or more examples of the technology. The headings provided herein are for convenience only and are not intended to limit or interpret the scope or meaning of the claimed technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the drawings, like reference characters generally refer to the same parts throughout the different views. Also, the drawings are not necessarily to scale, with an emphasis instead generally being placed upon illustrating the principles of the invention. In the following description, various embodiments of the present invention are described with reference to the following drawings, in which:
- (2) FIG. 1 is a perspective view of a monitored workspace in accordance with an embodiment of the invention.
- (3) FIG. 2 schematically illustrates classification of regions within the monitored workspace in accordance with an embodiment of the invention.
- (4) FIG. 3 schematically illustrates a control system in accordance with an embodiment of the invention.
- (5) FIGS. 4A-4C schematically illustrate dynamically defined safe zones and unsafe zones in a workspace at three instants of a task in accordance with an embodiment of the invention.
- (6) FIG. 4D schematically depicts a constrained motion plan in accordance with an embodiment of the invention.
- (7) FIG. 5A schematically illustrates dynamically defined safe zones and unsafe zones in a workspace having a six-axis robot in accordance with an embodiment of the invention.
- (8) FIG. 5B schematically depicts a constrained motion plan in accordance with an embodiment of the invention.
- (9) FIG. 6 schematically illustrates an object-monitoring system in accordance with an embodiment of the invention.
- (10) FIG. 7 schematically illustrates the definition of progressive safety envelopes in proximity to a piece of industrial machinery.
- (11) FIG. 8 schematically illustrates dynamic adjustment of a motion plan in real-time in response to an updated safe/unsafe zone in accordance with an embodiment of the invention.

DETAILED DESCRIPTION

(12) In the following discussion, we describe an integrated system for monitoring a workspace, classifying regions therein for safety purposes, and dynamically identifying safe states. In some cases the latter function involves semantic analysis of a robot in the workspace and identification of the workpieces with which it interacts. It should be understood, however, that these various elements may be implemented separately or together in desired combinations; the inventive aspects discussed herein do not require all of the described elements, which are set forth together merely for case of presentation and to illustrate their interoperability. The system as described represents merely one embodiment.

(13) 1. Workspace Monitoring

(14) Refer first to FIG. 1, which illustrates a representative 3D workspace **100** monitored by a plurality of sensors representatively indicated at **1021**, **1022**, **1023**. The sensors **102** may be conventional optical sensors such as cameras, e.g., 3D time-of-flight cameras, stereo vision cameras, or 3D LIDAR sensors or radar-based sensors, ideally with high frame rates (e.g., between 20 Hz and 100 Hz). The mode of operation of the sensors **102** is not critical so long as a 3D representation of the workspace **100** is obtainable from images or other data obtained by the sensors **102**. As shown in the figure, sensors **102** collectively cover and can monitor the workspace **100**, which includes a robot **106** controlled by a conventional robot controller **108**. The robot interacts with various workpieces **W**, and a person **P** in the workspace **100** may interact with the workpieces and the robot **108**. The workspace **100** may also contain various items of auxiliary equipment **110**, which can complicate analysis of the workspace by occluding various portions thereof from the sensors. Indeed, any realistic arrangement of sensors will frequently be unable to “see” at least some portion of an active workspace. This is illustrated in the simplified arrangement of FIG. 1: due to the presence of the person **P**, at least some portion of robot controller **108** may be occluded from all sensors. In an environment that people traverse and where even stationary objects may be moved from time to time, the unobservable regions will shift and vary.

(15) As shown in FIG. 2, embodiments of the present invention classify workspace regions as occupied, unoccupied (or empty), or unknown. For ease of illustration, FIG. 2 shows two sensors **2021**, **2022** and their zones of coverage **2051**, **2052** within the workspace **200** in two dimensions; similarly, only the two-dimensional (2D) footprint **210** of a 3D object is shown. The portions of the coverage zones **205** between the object boundary and the sensors **200** are marked as unoccupied, because each sensor affirmatively detects no obstructions in this intervening space. The space at the object boundary is marked as occupied. In a coverage zone **205** beyond an object boundary, all space is marked as unknown; the corresponding sensor is configured to sense occupancy in this region but, because of the intervening object **210**, cannot do so.

(16) With renewed reference to FIG. 1, data from each sensor **102** is received by a control system **112**. The volume of space covered by each sensor typically a solid cone may be represented in any suitable fashion, e.g., the space may be divided into a 3D grid of small (5 cm, for example) cubes or “voxels” or other suitable form of volumetric representation. For example, workspace **100** may be represented using 2D or 3D ray tracing, where the intersections of the 2D or 3D rays emanating from the sensors **102** are used as the volume coordinates of the workspace **100**. This ray tracing can be performed dynamically or via the use of precomputed volumes, where objects in the workspace **100** are previously identified and captured by control system **112**. For convenience of presentation, the ensuing discussion assumes a voxel representation; control system **112** maintains an internal representation of the workspace **100** at the voxel level, with voxels marked as occupied, unoccupied, or unknown.

(17) FIG. 3 illustrates, in greater detail, a representative embodiment of control system **112**, which may be implemented on a general-purpose

computer and a specialized computing device (see, e.g., U.S. Patent App. Pub. No. 2020/0272123, the entire content of which is incorporated herein by reference). The control system **112** includes a central processing unit (CPU) **305**, system memory **310**, and one or more non-volatile mass storage devices (such as one or more hard disks and/or optical storage units) **312**. The system **112** further includes a bidirectional system bus **315** over which the CPU **305**, memory **310**, and storage device **312** communicate with each other as well as with internal or external input/output (I/O) devices such as a display **320** and peripherals **322**, which may include traditional input devices such as a keyboard or a mouse). The control system **112** also includes a wireless transceiver **325** and one or more I/O ports **327**. Transceiver **325** and I/O ports **327** may provide a network interface. The term “network” is herein used broadly to connote wired or wireless networks of computers or telecommunications devices (such as wired or wireless telephones, tablets, etc.). For example, a computer network may be a local area network (LAN) or a wide area network (WAN). When used in a LAN networking environment, computers may be connected to the LAN through a network interface or adapter; for example, a supervisor may establish communication with control system **112** using a tablet that wirelessly joins the network. When used in a WAN networking environment, computers typically include a modem or other communication mechanism. Modems may be internal or external, and may be connected to the system bus via the user-input interface, or other appropriate mechanism. Networked computers may be connected over the Internet, an Intranet, Extranet, Ethernet, or any other system that provides communications. Some suitable communications protocols include TCP/IP, UDP, or OSI, for example. For wireless communications, communications protocols may include IEEE 802.11x (“Wi-Fi”), BLUETOOTH, ZigBee, IrDa, near-field communication (NFC), or other suitable protocol. Furthermore, components of the system may communicate through a combination of wired or wireless paths, and communication may involve both computer and telecommunications networks.

(18) CPU **305** is typically a microprocessor, but in various embodiments may be a microcontroller, peripheral integrated circuit element, a CSIC (customer-specific integrated circuit), an ASIC (application-specific integrated circuit), a logic circuit, a digital signal processor, a programmable logic device such as an FPGA (field-programmable gate array), PLD (programmable logic device), PLA (programmable logic array), radiofrequency identification (RFID) processor, graphics processing unit (GPU), smart chip, or any other device or arrangement of devices that is capable of implementing the steps of the processes of the invention.

(19) The system memory **310** contains a series of frame buffers **335**, i.e., partitions that store, in digital form (e.g., as pixels or voxels, or as depth maps), images obtained by the sensors **102**; the data may actually arrive via I/O ports **327** and/or transceiver **325** as discussed above. System memory **310** contains instructions, conceptually illustrated as a group of modules, that control the operation of CPU **305** and its interaction with the other hardware components. An operating system **340** (e.g., Windows or Linux) directs the execution of low-level, basic system functions such as memory allocation, file management and operation of mass storage device **312**. At a higher level, and as described in greater detail below, an analysis module **342** registers the images in frame buffers **335** and analyzes them to classify regions of the monitored workspace **100**. The result of the classification may be stored in a space map **345**, which contains a volumetric representation of the workspace **100** with each voxel (or other unit of representation) labeled, within the space map, as described herein. Alternatively, space map **345** may simply be a 3D array of voxels, with voxel labels being stored in a separate database (in memory **310** or in mass storage **312**).

(20) In various embodiments, a motion plan-generation module **346** receives (i) a task specification from system memory **310** or mass storage **312** and (ii) data related to the classified regions of the monitored workspace from the analysis module **342** and/or space map **345**, and, based thereon, generates a constrained motion plan **348** according to which the robot **106** performs the specified task without entering the occupied or unknown volumes. The motion plan **348** may be stored in system memory **310**, mass storage **312** and/or suitable hardware carried out by software executing on the control system **112**. In one embodiment, the motion plan **348** includes more than one trajectory of the machinery for performing the task—i.e., all of the included trajectories can perform the task within the safety constraint(s) but involve different movement sequences. The system memory **310** further includes a selection module **349** for selecting an optimal trajectory among those identified in the motion plan. For example, selection module **349** may assign a cost value to each candidate trajectory based on, for example, the length of the trajectory, the operating time required for the machinery to execute the trajectory, or kinematic limitations on robot motion that make particular trajectories more difficult to execute. The selection module **349** may then select the optimal trajectory based on the cost value (e.g., the minimum cost corresponding to the shortest trajectory length, shortest operating time, or the lowest value of a metric capturing the kinematic limitations), thereby ensuring efficient and safe operation of the machinery.

(21) Control system **112** may also control the operation or machinery in the workspace **100** using conventional control routines collectively indicated at **350**. In one embodiment, the control routines are based on the generated motion plan **348** (and/or the selected trajectory of the motion plan). As explained below, the configuration of the workspace and, consequently, the classifications associated with its voxel representation may well change over time as persons and/or machines move about, and control routines **350** may be responsive to these changes in operating machinery to achieve high levels of safety. All of the modules in system memory **310** may be programmed in any suitable programming language, including, without limitation, high-level languages such as C, C++, C#, Ada, Basic, Cobra, Fortran, Java, Lisp, Perl, Python, Ruby, or low-level assembly languages.

(22) 1.1 Sensor Registration

(23) In a typical multi-sensor system, the precise location of each sensor **102** with respect to all other sensors is established during setup. Sensor registration is usually performed automatically, and should be as simple as possible to allow for case of setup and reconfiguration. Assuming for simplicity that each frame buffer **335** stores an image (which may be refreshed periodically) from a particular sensor **102**, analysis module **342** may register sensors **102** by comparing all or part of the image from each sensor to the images from other sensors in frame buffers **335**, and using conventional computer-vision techniques to identify correspondences in those images. Suitable global-registration algorithms, which do not require an initial registration approximation, generally fall into two categories: feature-based methods and intensity-based methods. Feature-based methods identify correspondences between image features such as edges while intensity-based methods use correlation metrics between intensity patterns. Once an approximate registration is identified, an Iterative Closest Point (ICP) algorithm or suitable variant thereof may be used to fine-tune the registration.

(24) If there is sufficient overlap between the fields of view of the various sensors **102**, and sufficient detail in the workspace **100** to provide distinct sensor images, it may be sufficient to compare images of the static workspace. If this is not the case, a “registration object” having a distinctive signature in 3D can be placed in a location within workspace **100** where it can be seen by all sensors. Alternatively, registration can be achieved by having the sensors **102** record images of one or more people standing in the workspace or walking throughout the workspace over a period of time, combining a sufficient number of partially matching images until accurate registration is achieved.

(25) Registration to machinery within the workspace **100** can, in some cases, be achieved without any additional instrumentation, especially if the machinery has a distinctive 3D shape (for example, a robot arm), so long as the machinery is visible to at least one sensor registered with respect to the others. Alternatively, a registration object can be used, or a user interface, shown in display **320** and displaying the scene observed by the sensors, may allow a user to designate certain parts of the image as key elements of the machinery under control. In some embodiments, the interface provides an interactive 3D display that shows the coverage of all sensors to aid in configuration. If the system is configured with some degree of high-level information about the machinery being controlled (for purposes of control routines **350**, for example)—such as the location(s) of dangerous part or parts of the machinery and the stopping time and/or distance-analysis module **342** may be configured to provide intelligent feedback as to whether the sensors are providing sufficient coverage and suggest placement for additional sensors.

(26) For example, analysis module **342** can be programmed to determine the minimum distance from the observed machinery at which it must detect a person in order to stop the machinery by the time the person reaches it (or a safety zone around it), given conservative estimates of walking speed. (Alternatively, the required detection distance can be input directly into the system via display **320**.) Optionally, analysis module **342** can then analyze the fields of view of all sensors to determine whether the space is sufficiently covered to detect all approaches. If the sensor coverage is

insufficient analysis module **342** can propose new locations for existing sensors, or locations for additional sensors, that would remedy the deficiency. Otherwise, the control system will default to a safe state and control routines **350** will not permit machinery to operate unless analysis module **342** verifies that all approaches can be monitored effectively. Use of machine learning and genetic or evolutionary algorithms can be used to determine optimal sensor placement within a cell. Parameters to optimize include but are not limited to minimizing occlusions around the robot during operation and observability of the robot and workpieces.

(27) If desired, this static analysis may include “background” subtraction. During an initial startup period, when it may be safely assumed there no objects intruding into the workspace **100**, analysis module **342** identifies all voxels occupied by the static elements. Those elements can then be subtracted from future measurements and not considered as potential intruding objects. Nonetheless, continuous monitoring is performed to ensure that the observed background image is consistent with the space map **345** stored during the startup period. Background can also be updated if stationary objects are removed or are added to the workspace.

(28) There may be some areas that sensors **102** cannot observe sufficiently to provide safety, but that are guarded by other methods such as cages, etc. In this case, the user interface can allow the user to designate these areas as safe, overriding the sensor-based safety analysis. Safety-rated soft-axis and rate limitations can also be used to limit the envelope of the robot to improve performance of the system.

(29) Once registration has been achieved, sensors **102** should remain in the same location and orientation while the workspace **100** is monitored. If one or more sensors **102** are accidentally moved, the resulting control outputs will be invalid and could result in a safety hazard. Analysis module **342** may extend the algorithms used for initial registration to monitor continued accuracy of registration. For example, during initial registration analysis module **342** may compute a metric capturing the accuracy of fit of the observed data to a model of the work cell static elements that is captured during the registration process. As the system operates, the same metric can be recalculated. If at any time that metric exceeds a specified threshold, the registration is considered to be invalid and an error condition is triggered; in response, if any machinery is operating, a control routine **350** may halt it or transition the machinery to a safe state.

(30) 1.2 Identifying Occupied and Potentially Occupied Areas

(31) Once the sensors have been registered, control system **112** periodically updates space map **345** at a high fixed frequency (e.g., every analysis cycle) in order to be able to identify all intrusions into workspace **100**. Space map **345** reflects a fusion of data from some or all of the sensors **102**. But given the nature of 3D data, depending on the locations of the sensors **102** and the configuration of workspace **100**, it is possible that an object in one location will occlude the sensor's view of objects in other locations, including objects (which may include people or parts of people, e.g., arms) that are closer to the dangerous machinery than the occluding object. Therefore, to provide a reliably safe system, the system monitors occluded space as well as occupied space.

(32) In one embodiment, space map **345** is a voxel grid. In general, each voxel may be marked as occupied, unoccupied or unknown; only empty space can ultimately be considered safe, and only when any additional safety criteria—e.g., minimum distance from a piece of controlled machinery—is satisfied. Raw data from each sensor is analyzed to determine whether, for each voxel, an object or boundary of the 3D mapped space has been definitively detected in the volume corresponding to that voxel. To enhance safety, analysis module **342** may designate as empty only voxels that are observed to be empty by more than one sensor **102**. Again, all space that cannot be confirmed as empty is marked as unknown. Thus, only space between a sensor **102** and a detected object or mapped 3D space boundary along a ray may be marked as empty.

(33) If a sensor detects anything in a given voxel, all voxels that lie on the ray beginning at the focal point of that sensor and passing through the occupied voxel, and which are between the focal point and the occupied voxel, are classified as unoccupied, while all voxels that lie beyond the occupied voxel on that ray are classified as occluded for that sensor; all such occluded voxels are considered “unknown.” Information from all sensors may be combined to determine which areas are occluded from all sensors; these areas are considered unknown and therefore unsafe. Analysis module **342** may finally mark as “unoccupied” only voxels or workspace volumes that have been preliminarily marked at least once (or, in some embodiments, at least twice) as “unoccupied.” Based on the markings associated with the voxels or discrete volumes within the workspace, analysis module **342** may map one or more safe volumetric zones within space map **345**. These safe zones are outside a safety zone of the machinery and include only voxels or workspace volumes marked as unoccupied.

(34) A common failure mode of active optical sensors that depend on reflection, such as LIDAR and time-of-flight cameras, is that they do not return any signal from surfaces that are insufficiently reflective, and/or when the angle of incidence between the sensor and the surface is too shallow. This can lead to a dangerous failure because this signal can be indistinguishable from the result that is returned if no obstacle is encountered; the sensor, in other words, will report an empty voxel despite the possible presence of an obstacle. This is why ISO standards for e.g., 2D LIDAR sensors have specifications for the minimum reflectivity of objects that must be detected; however, these reflectivity standards can be difficult to meet for some 3D sensor modalities such as time-of-flight (TOF). In order to mitigate this failure mode, analysis module **342** marks space as empty only if some obstacle is definitively detected at further range along the same ray. By pointing sensors slightly downward so that most of the rays will encounter the floor if no obstacles are present, it is possible to conclusively analyze most of the workspace **100**. But if the sensed light level in a given voxel is insufficient to definitively establish emptiness or the presence of a boundary, the voxel is marked as unknown. The signal and threshold value may depend on the type of sensor being used. In the case of an intensity-based 3D sensor (for example, a time-of-flight camera) the threshold value can be a signal intensity, which may be attenuated by objects in the workspace of low reflectivity. In the case of a stereo vision system, the threshold may be the ability to resolve individual objects in the field of view. Other signal and threshold value combinations can be utilized depending on the type of sensor used.

(35) A safe system can be created by treating all unknown space as though it were occupied. However, in some cases this may be overly conservative and result in poor performance. It is therefore desirable to further classify unknown space according to whether it could potentially be occupied. As a person moves within a 3D space, he or she will typically occlude some areas from some sensors, resulting in areas of space that are temporarily unknown (see FIG. 1). Additionally, moving machinery such as an industrial robot arm can also temporarily occlude some areas. When the person or machinery moves to a different location, one or more sensors will once again be able to observe the unknown space and return it to the confirmed-empty state in which it is safe for the robot or machine to operate. Accordingly, in some embodiments, space may also be classified as “potentially occupied.” Unknown space is considered potentially occupied when a condition arises where unknown space could be occupied. This could occur when unknown space is adjacent to entry points to the workspace or if unknown space is adjacent to occupied or potentially occupied space. The potentially occupied space “infects” unknown space at a rate that is representative of a human moving through the workspace. Potentially occupied space stays potentially occupied until it is observed to be empty. For safety purposes, potentially occupied space is treated the same as occupied space. It may be desirable to use probabilistic techniques such as those based on Bayesian filtering to determine the state of each voxel, allowing the system to combine data from multiple samples to provide higher levels of confidence in the results. Suitable models of human movement, including predicted speeds (e.g., an arm may be raised faster than a person can walk), are readily available.

(36) 2. Classifying Objects

(37) For many applications, the classification of regions in a workspace as described above may be sufficient—e.g., if control system **112** is monitoring space in which there should be no objects at all during normal operation. In many cases, however, it is desirable to monitor an area in which there are at least some objects during normal operation, such as one or more machines and workpieces on which the machine is operating. In these cases, analysis module **342** may be configured to identify intruding objects that are unexpected or that may be humans. One suitable approach to such classification is to cluster individual occupied voxels into objects that can be analyzed at a higher level.

(38) To achieve this, analysis module **342** may implement any of several conventional, well-known clustering techniques such as Euclidean clustering, K-means clustering and Gibbs-sampling clustering. Any of these or similar algorithms can be used to identify clusters of occupied voxels

from 3D point cloud data. Mesh techniques, which determine a mesh that best fits the point-cloud data and then use the mesh shape to determine optimal clustering, may also be used. Once identified, these clusters can be useful in various ways.

(39) One simple way clustering can be used is to eliminate small groups of occupied or potentially occupied voxels that are too small to possibly contain a person. Such small clusters may arise from occupation and occlusion analysis, as described above, and can otherwise cause control system **112** to incorrectly identify a hazard. Clusters can be tracked over time by simply associating identified clusters in each image frame with nearby clusters in previous frames or using more sophisticated image-processing techniques. The shape, size, or other features of a cluster can be identified and tracked from one frame to the next. Such features can be used to confirm associations between clusters from frame to frame, or to identify the motion of a cluster. This information can be used to enhance or enable some of the classification techniques described below. Additionally, tracking clusters of points can be employed to identify incorrect and thus potentially hazardous situations. For example, a cluster that was not present in previous frames and is not close to a known border of the field of view may indicate an error condition.

(40) In some cases it may be sufficient to filter out clusters below a certain size and to identify cluster transitions that indicate error states. In other cases, however, it may be necessary to further classify objects into one or more of four categories: (1) elements of the machinery being controlled by system **112**, (2) the workpiece or workpieces that the machinery is operating on, and (3) other foreign objects, including people, that may be moving in unpredictable ways and that can be harmed by the machinery. It may or may not be necessary to conclusively classify people versus other unknown foreign objects. It may be necessary to definitively identify elements of the machinery as such, because by definition these will always be in a state of “collision” with the machinery itself and thus will cause the system to erroneously stop the machinery if detected and not properly classified. Similarly, machinery typically comes into contact with workpieces, but it is typically hazardous for machinery to come into contact with people. Therefore, analysis module **342** should be able to distinguish between workpieces and unknown foreign objects, especially people.

(41) Elements of the machinery itself may be handled for classification purposes by the optional background-subtraction calibration step described above. In cases where the machinery changes shape, elements of the machinery can be identified and classified, e.g., by supplying analysis module **342** with information about these elements (e.g., as scalable 3D representations), and in some cases (such as industrial robot arms) providing a source of instantaneous information about the state of the machinery. Analysis module **342** may be “trained” by operating machinery, conveyors, etc. in isolation under observation by the sensors **102**, allowing analysis module **342** to learn their precise regions of operation resulting from execution of the full repertoire of motions and poses. Analysis module **342** may classify the resulting spatial regions as occupied.

(42) Conventional computer-vision techniques may be employed to enable analysis module **342** to distinguish between workpieces and humans. These include deep learning, a branch of machine learning designed to use higher levels of abstraction in data. The most successful of these deep-learning algorithms have been convolutional neural networks (CNNs) and more recently recurrent neural networks (RNNs). However, such techniques are generally employed in situations where accidental misidentification of a human as a non-human does not cause safety hazards. In order to use such techniques in the present environment, a number of modifications may be needed. First, machine-learning algorithms can generally be tuned to prefer false positives or false negatives (for example, logistic regression can be tuned for high specificity and low sensitivity). False positives in this scenario do not create a safety hazard—if the robot mistakes a workpiece for a human, it will react conservatively. Additionally, multiple algorithms or neural networks based on different image properties can be used, promoting the diversity that may be key to achieving sufficient reliability for safety ratings. One particularly valuable source of diversity can be obtained by using sensors that provide both 3D and 2D image data of the same object. If any one technique identifies an object as human, the object will be treated as human. Using multiple techniques or machine-learning algorithms, all tuned to favor false positives over false negatives, sufficient reliability can be achieved. In addition, multiple images can be tracked over time, further enhancing reliability—and again every object can be treated as human until enough identifications have characterized it as non-human to achieve reliability metrics. Essentially, this diverse algorithmic approach, rather than identifying humans, identifies things that are definitely not humans.

(43) In addition to combining classification techniques, it is possible to identify workpieces in ways that do not rely on any type of human classification at all. One approach is to configure the system by providing models of workpieces. For example, a “teaching” step in system configuration may simply supply images or key features of a workpiece to analysis module **342**, which searches for matching configurations in space map **345**, or may instead involve training of a neural network to automatically classify workpieces as such in the space map. In either case, only objects that accurately match the stored model are treated as workpieces, while all other objects are treated as humans.

(44) Another suitable approach is to specify particular regions within the workspace, as represented in the space map **345**, where workpieces will enter (such as the top of a conveyor belt). Only objects that enter the workspace in that location are eligible for treatment as workpieces. The workpieces can then be modeled and tracked from the time they enter the workspace until the time they leave. While a monitored machine such as a robot is handling a workpiece, control system **112** ensures that the workpiece is moving only in a manner consistent with the expected motion of the robot end effector. Known equipment such as conveyor belts can also be modeled in this manner. Humans may be forbidden from entering the work cell in the manner of a workpiece—e.g., sitting on conveyors.

(45) All of these techniques can be used separately or in combination, depending on design requirements and environmental constraints. In all cases, however, there may be situations where analysis module **342** loses track of whether an identified object is a workpiece. In these situations the system should fall back to a safe state. An interlock can then be placed in a safe area of the workspace where a human worker can confirm that no foreign objects are present, allowing the system to resume operation.

(46) In some situations a foreign object enters the workspace, but subsequently should be ignored or treated as a workpiece. For example, a stack of boxes that was not present in the workspace at configuration time may subsequently be placed therein. This type of situation, which will become more common as flexible systems replace fixed guarding, may be addressed by providing a user interface (e.g., shown in display **320** or on a device in wireless communication with control system **112**) that allows a human worker to designate the new object as safe for future interaction. Of course, analysis module **342** and control routines **350** may still act to prevent the machinery from colliding with the new object, but the new object will not be treated as a potentially human object that could move towards the machinery, thus allowing the system to handle it in a less conservative manner.

(47) 3. Generating a Constrained Motion Plan of Machinery

(48) At this stage, analysis module **342** has identified all objects in the monitored area **100** that must be considered for safety purposes. A constrained motion plan may now be generated for the machinery to perform a task or portion thereof without collision or violation of speed-and-separation monitoring (SSM) requirements as defined by the ISO 10218 and ISO/TS 15066 standards. For example, with reference to FIGS. **4A-4C**, assuming the robot **402** is required to move from point A, in front of the parts bin **404**, to point B, at the assembly table **406**, in executing a task, based on the task specification and the identified human operator **408** (which can be multiple humans around the monitored space **420**) and objects **410-416** in the monitored space **420**, analysis module **342** may classify space in the workspace **420** into safe zones (e.g., corresponding to unoccupied space) **424.sub.1-3** and unsafe zones (e.g., corresponding to occupied and/or unknown space) **426.sub.1-3** at three instants, t.sub.1-3, respectively, during execution of the task. Referring to FIG. **4D**, based on the mapped safe and unsafe zones, the motion plan-generation module **346** generates a constrained motion plan for the robot **106** including three trajectories **432-436** and/or one or more states (e.g., orientations, velocities, etc.) to perform the task without entering the unsafe zones **426.sub.1-3** in any of the instances.

(49) Because a motion plan **348** may include multiple trajectories, selection module **349** selects the optimal trajectory among those available in the motion plan. For example, selection module **349** may assign a cost value to each trajectory based on, for example, the path length of the trajectory and/or the time it will take robot **106** to traverse it, as well as any other quantifiable benefits or disadvantages to each of the paths (e.g., the desirability to keeping some areas of the workspace clear when possible, the proximity each path to entry points that increase the likelihood that a trajectory may become unsafe due to entry of a human, etc.). Based on the assigned cost value, selection module **349** selects the optimal trajectory

(e.g., having the minimum cost value), thereby ensuring efficient operation of the machinery. Subsequently, robot **106** can be operated based on the selected trajectory in the constrained motion plan as further described below.

(50) The approach described above may be implemented to generate and/or execute a constrained motion plan for part of the machinery, such as a robotic arm. Here the motion plan-generation module **346** may take joint kinematics and physical constraints into account when generating the motion plan. FIG. 5A depicts a six-axis articulated robot **502**; each joint of the robot **502** has its corresponding constrained kinematics. For example, the base joint **504** can only turn along its axis perpendicular to the floor (i.e., in the direction **506**); similarly, the joint **508** physically anchored to the base joint **504** may only move by rotating around a single axis (in the direction **510**) and have a pose in space constrained by the base joint's rotation; and so on. Assuming the robot's appendage **512** is required to move an object from point A to point B in executing a task, based on the task specification and an identified operator's arm **516** in the space monitored by the sensor system **102**, analysis module **342** may classify space in the workspace into safe zones **522** and unsafe zones **524** during execution of the task. With reference to FIG. 5B, based on the mapped safe and unsafe zones, the motion plan-generation module **346** generates a constrained motion plan including two trajectories **526**, **528** for performing the task without entering the unsafe zones **524** and risking collision with the operator's arm **516**.

(51) To identify the optimal trajectory for the robot **502**, selection module **349** may assign a cost value to each trajectory based on, for example, the path length of the trajectory, the time it will take the robot **502** to traverse it, allowed or physically possible joint kinematics through the trajectory, as well as any other quantifiable benefits or disadvantages to each of the paths. Again, based on the assigned cost value, selection module **349** selects the optimal trajectory (e.g., having the minimum cost value) and the robot can be operated based on the selected trajectory.

(52) 4. Generating Control Outputs

(53) Once analysis module **342** has identified all objects in the monitored area **100** that must be considered for safety purposes and/or motion plan-generation module **346** has generated a constrained motion plan, a variety of actions can be taken and control outputs generated. For example, during static calibration or with the workspace in a default configuration free of humans, space map **345** may be useful to a human for evaluating sensor coverage, the configuration of deployed machinery, and opportunities for unwanted interaction between humans and machines. Even without setting up cages or fixed guards, the overall workspace layout may be improved by channeling or encouraging human movement through the regions marked as safe zones, as described above, and away from regions with poor sensor coverage.

(54) Control routines **350**, responsive to analysis module **342**, may generate control signals to operating machinery, such as robots, within workspace **100** when certain conditions are detected. This control can be binary, indicating either safe or unsafe conditions, or can be more complex, such as an indication of what actions are safe and unsafe. The simplest type of control signal is a binary signal indicating whether an intrusion of either occupied or potentially occupied volume is detected in a particular zone. In the simplest case, there is a single intrusion zone and control system **112** provides a single output indicative of an intrusion. This output can be delivered, for example, via an I/O port **327** to a complementary port on the controlled machinery to stop or limit the operation of the machinery. In more complex scenarios, multiple zones are monitored separately, and a control routine **350** issues a digital output via an I/O port **327** or transceiver **325** addressed, over a network, to a target piece of machinery (e.g., using the Internet protocol or other suitable addressing scheme).

(55) During robot operation, control routines **350** generate control signals to execute a task following selection of a task-execution trajectory by selection module **349**. In particular, control routines **350** generate and deliver an output signal via, for example, the transceiver **325** and/or I/O port **327** to operate the machinery in accordance with the selected trajectory. As explained in greater detail below, the configuration of the safe zones around which the trajectory was planned may change during task execution—for example, a human may intrude into or close to a portion of the trajectory. Therefore, as noted, one element of the attributed cost of a trajectory may be the probability of a change in the safe-zone configuration required for the trajectory—e.g., the relative likelihood among trajectories that a human will appear and alter the safe zone.

(56) Another condition that may be monitored is the distance between any object in the workspace and a machine, comparable to the output of a 2D proximity sensor. This may be converted into a binary output by establishing a proximity threshold below which the output should be asserted. It may also be desirable for the system to record and make available the location and extent of the object closest to the machine. In other applications, such as a safety system for a collaborative industrial robot, the desired control output may include the location, shape, and extent of all objects observed within the area covered by the sensors **102**.

(57) 5. Safe Action Constraints and Dynamic Determination of Safe Zones

(58) ISO 10218 and ISO/TS 15066 describe speed and separation monitoring as a safety function that can enable collaboration between an industrial robot and a human worker. Risk reduction is achieved by maintaining at least a protective separation distance between the human worker and robot during periods of robot motion. This protective separation distance is calculated using information including robot and human worker position and movement, robot stopping distance, measurement uncertainty, system latency and system control frequency. When the calculated separation distance decreases to a value below the protective separation distance, the robot system is slowed down or stopped. This methodology can be generalized beyond industrial robotics to machinery.

(59) For convenience, the following discussion focuses on dynamically defining a safe zone around a robot operating in the workspace **100**. It should be understood, however, that the techniques described herein apply not only to multiple robots but to any form of machinery that can be dangerous when approached too closely, and which has a minimum safe separation distance that may vary over time and with particular activities undertaken by the machine. As described above, a sensor array obtains sufficient image information to characterize, in 3D, the robot and the location and extent of all relevant objects in the area surrounding the robot at each analysis cycle. (Each analysis cycle includes image capture, refresh of the frame buffers, and computational analysis; accordingly, although the period of the analysis or control cycle is short enough for effective monitoring to occur in real time, it involves many computer clock cycles.) Analysis module **342** utilizes this information along with instantaneous information about the current state of the robot at each cycle to determine instantaneous, current safe action constraints for the robot's motion. The constraints may be communicated to the robot, either directly by analysis module **342** or via a control routine **350**, to the robot via transceiver **325** or and I/O port **327**.

(60) Additionally, motion plan-generation module **346** may be responsive to real-time monitoring of the workspace and alter the motion plan in response to a change in the defined safe zone. For example, analysis module **342** frequently updates the defined safe zone based on the data acquired by the sensor array. Motion plan-generation module **346** may receive the updated safe zone definition from the analysis module **342** (or, in some embodiments, the space map **345**) as it is generated and, based thereon, adjust the motion plan such that the machinery will not enter the updated unsafe zone. Again, if the updated motion plan includes multiple trajectories of the machinery and the task has not yet been started, selection module **349** may update the cost values assigned thereto and select the optimal trajectory based on the updated cost values so as to ensure efficient operation of the machinery. If the task is in progress, multiple actions are possible. If the robot **106** is well into the task or if the configuration of the safe zone is changing rapidly, control routines **350** may simply stop the robot and wait for a short period in case the disruption is momentary. If nothing has changed by the end of this period, the motion plan-generation module **346** may re-execute and attempt to generate viable task-completion trajectories, each having an associated cost as before, that are consistent with the reconfigured safe zone. If motion plan-generation module **346** succeeds, the least-cost trajectory is selected and executed as discussed above. If not, robot **106** may remain in place or, if possible consistent with the new safe zone, return to position and pose at the beginning of task execution. If task execution has just begun, control routines **350** may immediately reset the robot **106** to the pre-task position and pose, and determine whether the task can be completed once the new safe zone has been stable for a threshold time period, e.g., a few seconds.

(61) The operation of the system is best understood with reference to the conceptual illustration of system organization and operation of FIG. 6. As described above, a sensor array **102** monitors the workspace **600**, which includes a robot **602**. The robot's movements are controlled by a conventional robot controller **607**, which may be part of or separate from the robot itself; for example, a single robot controller may issue commands

to more than one robot. The robot's activities may primarily involve a robot arm, the movements of which are orchestrated by robot controller **607** using joint commands that operate the robot arm joints to effect a desired movement. An object-monitoring system (OMS) **610** obtains information about objects from the sensors **102** and uses this sensor information to identify relevant objects in the workspace **600**. OMS **610** communicates with robot controller **607** via any suitable wired or wireless protocol. (In an industrial robot, control electronics typically reside in an external control box. However, in the case of a robot with a built-in controller, OMS **610** communicates directly with the robot's onboard controller.) Using information obtained from the robot (and, typically, sensors **102**), OMS **610** determines the robot's current state. OMS **610** thereupon determines safe-action constraints, the constrained motion plan and/or the optimal trajectory for robot **602** given the robot's current state and all identified relevant objects. Finally, OMS **610** communicates the safe action constraints, constrained motion plan and/or optimal trajectory to robot **607**. (It will be appreciated that, with reference to FIG. 3, the functions of OMS **610** are performed in a control system **112** by analysis module **342**, motion-plan-generation module **346**, and selection module **349** and, in some cases, a control routine **350**.)

(62) 5.1 Identifying Relevant Objects

(63) The sensors **102** provide real-time image information that is analyzed by an object-analysis module **615** at a fixed frequency in the manner discussed above; in particular, at each cycle, object analysis module **615** identifies the precise 3D location and extent of all objects in workspace **600** that are either within the robot's reach or that could move into the robot's reach at conservative expected velocities. If not all of the relevant volume is within the collective field of view of the sensors **102**, OMS **610** may be configured to so determine and indicate the location and extent of all fixed objects within that region (or a conservative superset of those objects) and/or verify that other guarding techniques have been used to prevent access to unmonitored areas.

(64) 5.2 Determining Robot State

(65) A robot state determination module (RSDM) **620** is responsive to data from sensors **102** and signals from the robot **602** and/or robot controller **607** to determine the instantaneous state of the robot. In particular, RSDM **620** determines the pose and location of robot **602** within workspace **600**; this may be achieved using sensors **102**, signals from the robot and/or its controller, or data from some combination of these sources. RSDM **620** may also determine the instantaneous velocity of robot **602** or any appendage thereof; in addition, knowledge of the robot's instantaneous joint accelerations or torques, or planned future trajectory may be needed in order to determine safe motion constraints for the subsequent cycle as described below. Typically, this information comes from robot controller **607**, but in some cases may be inferred directly from images recorded by sensors **102** as described below.

(66) For example, these data could be provided by the robot **602** or the robot controller **607** via a safety-rated communication protocol providing access to safety-rated data. The 3D pose of the robot may then be determined by combining provided joint positions with a static 3D model of each link to obtain the 3D shape of the entire robot **602**.

(67) In some cases, the robot may provide an interface to obtain joint positions that is not safety-rated, in which case the joint positions can be verified against images from sensors **102** (using, for example, safety-rated software). For example, received joint positions may be combined with static 3D models of each link to generate a 3D model of the entire robot **602**. This 3D image can be used to remove any objects in the sensing data that are part of the robot itself. If the joint positions are correct, this will fully eliminate all object data attributed to the robot **602**. If, however, the joint positions are incorrect, the true position of robot **602** will diverge from the model, and some parts of the detected robot will not be removed. Those points will then appear as a foreign object in the new cycle. In the previous cycle, it can be assumed that the joint positions were correct because otherwise robot **602** would have been halted. Since the base joint of the robot does not move, at least one of the divergent points must be close to the robot. The detection of an unexpected object close to robot **602** can then be used to trigger an error condition, which will cause control system **112** (see FIG. 1) to transition robot **602** to a safe state. Alternately, sensor data can be used to identify the position of the robot using a correlation algorithm, such as described above in the section on registration, and this detected position can be compared with the joint position reported by the robot. If the joint position information provided by robot **602** has been validated in this manner, it can be used to validate joint velocity information, which can then be used to predict future joint positions. If these positions are inconsistent with previously validated actual joint positions, the program can similarly trigger an error condition. These techniques enable use of a non-safety-rated interface to produce data that can then be used to perform additional safety functions.

(68) Finally, RSDM **620** may be configured to determine the robot's joint state using only image information provided by sensors **102**, without any information provided by robot **602** or controller **607**. Given a model of all of the links in the robot, any of several conventional, well-known computer vision techniques can be used by RSDM **620** to register the model to sensor data, thus determining the location of the modeled object in the image. For example, the ICP algorithm (discussed above) minimizes the difference between two 3D point clouds. ICP often provides a locally optimal solution efficiently, and thus can be used accurately if the approximate location is already known. This will be the case if the algorithm is run every cycle, since robot **602** cannot have moved far from its previous position. Accordingly, globally optimal registration techniques, which may not be efficient enough to run in real time, are not required. Digital filters such as Kalman filters or particle filters can then be used to determine instantaneous joint velocities given the joint positions identified by the registration algorithm.

(69) These image-based monitoring techniques often rely on being run at each system cycle, and on the assumption that the system was in a safe state at the previous cycle. Therefore, a test may be executed when robot **602** is started—for example, confirming that the robot is in a known, pre-configured “home” position and that all joint velocities are zero. It is common for automated equipment to have a set of tests that are executed by an operator at a fixed interval, for example, when the equipment is started up or on shift changes. Reliable state analysis typically requires an accurate model of each robot link. This model can be obtained a priori, e.g. from 3D CAD files provided by the robot manufacturer or generated by industrial engineers for a specific project. However, such models may not be available, at least not for the robot and all of the possible attachments it may have.

(70) In this case, it is possible for RSDM **620** to create the model itself, e.g., using sensors **102**. This may be done in a separate training mode where robot **602** runs through a set of motions, e.g., the motions that are intended for use in the given application and/or a set of motions designed to provide sensors **102** with appropriate views of each link. It is possible, but not necessary, to provide some basic information about the robot a priori, such as the lengths and rotational axes of each link. During this training mode, RSDM **620** generates a 3D model of each link, complete with all necessary attachments. This model can then be used by RSDM **620** in conjunction with sensor images to determine the robot state.

(71) 5.3 Determining Safe-Action Constraints

(72) In traditional axis- and rate-limitation applications, an industrial engineer calculates what actions are safe for a robot, given the planned trajectory of the robot and the layout of the workspace forbidding some areas of the robot's range of motion altogether and limiting speed in other areas. These limits assume a fixed, static workplace environment. Here we are concerned with dynamic environments in which objects and people come, go, and change position; hence, safe actions are calculated by a safe-action determination/motion plan generation module **625** (SADM **625**) in real time based on all sensed relevant objects and on the current state of robot **602**, and these safe actions may be updated each cycle. In order to be considered safe, actions should ensure that robot **602** does not collide with any stationary object, and also that robot **602** does not come into contact with a person who may be moving toward the robot. Since robot **602** has some maximum possible deceleration, controller **607** should be instructed to begin slowing the robot down sufficiently in advance to ensure that it can reach a complete stop before contact is made. Additionally or alternatively, SADM **625** may instruct controller **607** to change the trajectory and/or pose of robot **602** based on the selected trajectory in the constrained motion plan, which is updated in real-time in accordance with the monitored workspace as described above—thereby avoiding collision with the person and other objects and ensuring safety.

(73) One approach to reducing the robot's velocity is to modulate the robot's maximum velocity (by which is meant the velocity of the robot itself or any appendage thereof) proportionally to the minimum distance between any point on the robot and any point in the relevant set of sensed objects to

be avoided. The robot is allowed to operate at maximum speed when the closest object is further away than some threshold distance beyond which collisions are not a concern, and the robot is halted altogether if an object is within a certain minimum distance. Sufficient margin can be added to the specified distances to account for movement of relevant objects or humans toward the robot at some maximum realistic velocity. This is illustrated in FIG. 7. An outer envelope or 3D zone **702** is generated computationally by SADM **625** around the robot **704**. Outside this zone **702**, all movements of the person P are considered safe because, within an operational cycle, they cannot bring the person sufficiently close to the robot **704** to pose a danger. Detection of any portion of the person P's body within a second 3D zone **708**, computationally defined within zone **702**, is registered by SADM **625** but robot **704** is allowed to continue operating at full speed. If any portion of the person P crosses the threshold of zone **708** but is still outside an interior danger zone **710**, robot **704** is signaled to operate at a slower speed. If any portion of the person P crosses into the danger zone **710**—or is predicted to do so within the next cycle based on a model of human movement—operation of robot **704** is halted. These zones may be updated if robot **704** is moved (or moves) within the environment.

(74) A refinement of this technique is for SADM **625** to control maximum velocity proportionally to the square root of the minimum distance, which reflects the fact that in a constant-deceleration scenario, velocity changes proportionally to the square root of the distance traveled, resulting in a smoother and more efficient, but still equally safe, result. A further refinement is for SADM **625** to modulate maximum velocity proportionally to the minimum possible time to collision—that is, to project the robot's current state forward in time, project the intrusions toward the robot trajectory, and identify the nearest potential collision. This refinement has the advantage that the robot will move more quickly away from an obstacle than toward it, which maximizes throughput while still correctly preserving safety. Since the robot's future trajectory depends not just on its current velocity but on subsequent commands, SADM **625** may consider all points reachable by robot **602** within a certain reaction time given its current joint positions and velocities, and cause control signals to be issued based on the minimum collision time among any of these states. Yet a further refinement is for SADM **625** to take into account the entire planned trajectory of the robot when making this calculation, rather than simply the instantaneous joint velocities.

(75) Additionally, SADM **625** may, via robot controller **607**, alter the robot's trajectory, rather than simply alter the maximum speed along that trajectory. It is possible to choose from among a fixed set of trajectories one that reduces or eliminates potential collisions, or even to generate a new trajectory on the fly. For example, referring to FIG. 8, robot **802** may be operated based on a selected trajectory **804** in a constrained motion plan generated at an instant t.sub.1. At instant t.sub.2, analysis module **342**, upon analyzing data received from sensor system **102**, detects an operator's arm in a region **806** and, based thereon, updates region **806** as an unsafe zone. In addition, the analysis module **342** may define the regions **808**, **810** as safe zones as the voxels associated therewith are marked as unoccupied. Based on the updated safe/unsafe zone definition from analysis module **342**, SADM **625** may adjust the motion plan in real-time such that robot **802** will not enter the updated unsafe zone **806**. Again, because the updated motion plan may include multiple trajectories **812**, **814** of robot **802** via going through the region **808** or **810**, respectively, the cost values assigned to trajectories **812**, **814** may be updated, and the optimal trajectory of the robot **802** may be selected based on the updated cost values as described above.

(76) While not necessarily a safety violation, collisions with static elements of the workspace are generally not desirable. The set of relevant objects can include all objects in the workspace, including both static background such as walls and tables, and moving objects such as workpieces and human workers. Either from prior configuration or run-time detection, sensors **102** and analysis module **342** may be able to infer which objects could possibly be moving. In this case, any of the algorithms described above can be refined to leave additional margins to account for objects that might be moving, but to eliminate those margins for objects that are known to be static, so as not to reduce throughput unnecessarily but still automatically eliminate the possibility of collisions with static parts of the work cell.

(77) Beyond simply leaving margins to account for the maximum velocity of potentially moving objects, state estimation techniques based on information detected by the sensing system can be used to project the movements of humans and other objects forward in time, thus expanding the control options available to control routines **350**. For example, skeletal tracking techniques can be used to identify moving limbs of humans that have been detected and limit potential collisions based on properties of the human body and estimated movements of, e.g., a person's arm rather than the entire person.

(78) 5.4 Communicating Safe Action Constraints to the Robot

(79) The safe-action constraints identified by SADM **625** may be communicated by OMS **610** to robot controller **607** on each cycle via a robot communication module **630**. As described above, the communication module may correspond to an I/O port **327** interface to a complementary port on robot controller **607** or may correspond to transceiver **325**. Most industrial robots provide a variety of interfaces for use with external devices. A suitable interface should operate with low latency at least at the control frequency of the system. The interface can be configured to allow the robot to be programmed and run as usual, with a maximum velocity, a trajectory and/or a pose being sent over the interface. In accordance with the inputs on the interface, SADM **625** may generate/update a constrained motion plan of the robot and, via the robot controller **607**, operate the robot based on the generated/updated motion plan as described above. Alternatively, some interfaces allow for trajectories to be delivered in the form of waypoints. Using this type of an interface, the intended trajectory of robot **602** can be received and stored within OMS **610**, which may then generate waypoints that are closer together or further apart depending on the safe-action constraints. Similarly, an interface that allows input of target joint torques can be used to drive trajectories computed in accordance herewith. These types of interface can also be used where SADM **625** chooses new trajectories or modifies trajectories depending on the safe-action constraints.

(80) As with the interface used to determine robot state, if robot **602** supports a safety-rated protocol that provides real-time access to the relevant safety-rated control inputs, this may be sufficient. However, a safety-rated protocol is not available, additional safety-rated software on the system can be used to ensure that the entire system remains safe. For example, SADM **625** may determine the expected speed and position of the robot if the robot is operating in accordance with the safe actions that have been communicated. SADM **625** then determines the robot's actual state as described above. If the robot's actions do not correspond to the expected actions, SADM **625** causes the robot to transition to a safe state, typically using an emergency stop signal. This effectively implements a real-time safety-rated control scheme without requiring a real-time safety-rated interface beyond a safety-rated stopping mechanism.

(81) In some cases a hybrid system may be optimal—many robots have a digital input that can be used to hold a safety-monitored stop. It may be desirable to use a communication protocol for variable speed, for example, when intruding objects are relatively far from the robot, but to use a digital safety-monitored stop when the robot must come to a complete stop, for example, when intruding objects are close to the robot.

(82) Certain embodiments of the present invention are described above. It is, however, expressly noted that the present invention is not limited to those embodiments; rather, additions and modifications to what is expressly described herein are also included within the scope of the invention.

Claims

1. A method of safely operating machinery in a workspace, the method comprising: recording images of a portion of a workspace; generating a 3D representation of the portion of the workspace based on the recorded images, wherein the 3D representation comprises one or more volumes that correspond to the portion of the workspace; identifying one or more of the volumes as being either occupied or unoccupied; and mapping one or more safe zones based on the one or more identified volumes, wherein the safe zones correspond to one or more regions within the portion of the workspace for safe operation of machinery.
2. The method of claim 1, wherein a particular safe zone of the one or more safe zones is detected based on user input.
3. The method of claim 2, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels

configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and the particular safe zone is disposed outside the sensor field of view.

4. The method of claim 2, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and the user input overrides a mapping based on images recorded by the sensor.

5. The method of claim 1, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and identifying a particular volume of the one or more volumes as occupied comprises receiving light via the sensor, wherein the light is reflected from an object disposed in the particular volume, and an intensity of the reflected light is higher than a predetermined threshold.

6. The method of claim 1, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and one or more objects within the sensor field of view are insufficiently reflective to reflect to the sensor light with an intensity higher than a predetermined threshold.

7. The method of claim 1, further comprising associating the one or more safe zones with a safety criterion.

8. The method of claim 7, wherein the safety criterion corresponds to a minimum distance from a piece of the machinery.

9. The method of claim 1, wherein the one or more safe zones include only volumes of the one or more volumes that have been identified as unoccupied.

10. The method of claim 1, further comprising, based on the recorded images, identifying a workpiece or an element of the machinery using a neural network.

11. The method of claim 1, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and identifying one or more of the volumes as being occupied or unoccupied comprises (i) detecting an occlusion in the sensor field of view, (ii) classifying volumes associated with the occlusion as occupied volumes, and (iii) classifying volumes between the sensor and the occupied volumes as unoccupied volumes.

12. The method of claim 1, wherein: recording images of the portion of the workspace comprises using a sensor associated with a grid of pixels configured to capture the recorded images, wherein the portion of the workspace is within a field of view of the sensor; and the sensor comprises at least one of (i) a time-of-flight sensor, (ii) a 3D LIDAR sensor, or (iii) a stereo vision camera.

13. The method of claim 1, further comprising: recording images of another portion of a workspace; generating another 3D representation of the other portion of the workspace based on the recorded images, wherein the other 3D representation comprises one or more other volumes that correspond to the other portion of the workspace; and identifying one or more of the other volumes as being either occupied or unoccupied.

14. The method of claim 1, further comprising: receiving a task specification; and generating a motion plan comprising a trajectory of the machinery for performing a specified task within the one or more safe zones.

15. The method of claim 14, further comprising: recording images of another portion of a workspace; generating another 3D representation of the other portion of the workspace based on the recorded images, wherein the other 3D representation comprises one or more other volumes that correspond to the other portion of the workspace; and identifying one or more of the other volumes as being either occupied or unoccupied; wherein mapping the one or more safe zones is further based on the one or more identified other volumes.

16. The method of claim 14, further comprising operating the machinery in accordance with the motion plan.

17. The method of claim 14, wherein: the motion plan comprises a plurality of trajectories of the machinery; and the method further comprises (i) assigning a respective cost value to each of the trajectories, (ii) selecting one of the trajectories based at least in part on the respective cost value assigned to each of the trajectories, and (iii) causing the machinery to execute the selected trajectory.

18. The method of claim 17, wherein the respective cost value associated with each of the trajectories is assigned based at least in part on (i) a length of the trajectory, (ii) an operation time of the machinery on the trajectory, or (iii) allowed kinematics of the trajectory.

19. The method of claim 14, wherein the motion plan precludes entry into a zone outside the one or more safe zones by the machinery and any workpiece associated with the machinery or the task.

20. The method of claim 14, further comprising: monitoring the workspace in real-time; and altering the motion plan in response to a change in the one or more safe zones.
